

National University of Computer and Emerging Sciences



Lab Manual # 6

Programming Fundamentals

(Section BCS- 1B)

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Section	BCS-1B
Semester	Fall 2022

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Lab Manual

Objectives

The objectives of this lab are to cover the following:

- for loop
- Arrays

Exercise 1:

Write a C++ program in which, take start and end as input from user. Display all perfect numbers (which are sum of its positive divisors except itself) between start and end.

Example:

Sample Input:

Enter start value: 1

Enter end value 10

Sample Output:

List perfect number 1 to 10 are: 6

//(As, $6 = 1+2+3$)

Exercise 2:

Write a program in C++ that prints first 10 entries of multiplication tables from 1 to 5

```
----- Table of 1 -----
1 x 1 : 1
1 x 2 : 2
1 x 3 : 3
1 x 4 : 4
1 x 5 : 5
1 x 6 : 6
1 x 7 : 7
1 x 8 : 8
1 x 9 : 9
1 x 10 : 10
-----
----- Table of 2 -----
2 x 1 : 2
2 x 2 : 4
2 x 3 : 6
2 x 4 : 8
2 x 5 : 10
2 x 6 : 12
2 x 7 : 14
2 x 8 : 16
2 x 9 : 18
2 x 10 : 20
-----
----- Table of 3 -----
3 x 1 : 3
3 x 2 : 6
3 x 3 : 9
3 x 4 : 12
3 x 5 : 15
3 x 6 : 18
3 x 7 : 21
3 x 8 : 24
3 x 9 : 27
3 x 10 : 30
-----
----- Table of 4 -----
4 x 1 : 4
4 x 2 : 8
4 x 3 : 12
4 x 4 : 16
4 x 5 : 20
4 x 6 : 24
4 x 7 : 28
4 x 8 : 32
4 x 9 : 36
4 x 10 : 40
-----
----- Table of 5 -----
5 x 1 : 5
5 x 2 : 10
5 x 3 : 15
5 x 4 : 20
5 x 5 : 25
5 x 6 : 30
5 x 7 : 35
5 x 8 : 40
5 x 9 : 45
5 x 10 : 50
-----
Press any key to continue . . .
```

Exercise 3:

Take Input the total number of lines from the user and then display the following equilateral triangle shape.

For example:

If the user inputs 3, then your program should print:

```
*  
***  
*****
```

Exercise 4:

Take Input the total number of lines from the user and then display the following equilateral triangle shape.

For example:

If the user inputs 3, then your program should print:

```
*****  
***  
*
```

Exercise 5:

Write a program in C++ that asks the user to enter size and displays a triangle of the required size e.g. if the the user enters 4 as size then a Triangle of size 4 will be displayed on the screen as shown below:

```
1  
1 2  
1 2 3  
1 2 3 4  
1 2 3  
1 2  
1
```

Exercise 6 (Arrays):

Write a program in C++ that declares a static integer array of size 5, then in a loop asks the user to enter 5 integer values and stores them in the array. Display the values stored in array.